

Rank-Deficient Precision Estimation in Unrolled GEM

Twicing for the E-step, Kalofolias-style Log-Barrier for the M-step

Setup: GEM with a GMRF Prior

- GMRF prior on latent signal $\mathbf{x}$:

$$\Pr(\mathbf{x} \mid \Theta) \propto \exp\left(-\frac{1}{2}\mathbf{x}^\top \Theta \mathbf{x}\right)$$

- Observation model: $\mathbf{y} = \mathbf{x} + \text{noise}$, $\Pr(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 I)$
- E-step: estimate $\mathbf{x}_t$ given current Θ_t
- M-step: update $\Theta_t \rightarrow \Theta_{t+1}$ from the E-step output

Problem: a single $\mathbf{x}_t$ gives empirical covariance

$$C_t = \mathbf{x}_t \mathbf{x}_t^\top \quad (\text{rank 1 — badly rank-deficient for precision estimation})$$

Two Sources of Diversity

(1) Historical iterates (temporal diversity)

$$\hat{C}_t^{\text{hist}} = \sum_k w_k \mathbf{x}_{t-k} \mathbf{x}_{t-k}^\top$$

Caution: $\mathbf{x}_{t-k}$ was produced under stale Θ_{t-k} — weights should decay with staleness.

(2) Deflation (spatial diversity, within one E-step)

$$\hat{C}_t^{\text{def}} = \sum_j v_j \mathbf{x}_t^{(j)} (\mathbf{x}_t^{(j)})^\top, \quad \mathbf{x}_t^{(j)} \perp \mathbf{x}_t$$

Hybrid:

$$C_t = \sum_i w_i \mathbf{x}_i \mathbf{x}_i^\top + \sum_j v_j \mathbf{x}_t^{(j)} (\mathbf{x}_t^{(j)})^\top, \quad \Theta_t = f_\phi(\mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_t^{(2)}, \dots)$$

with f_ϕ (mixture weights) learned end-to-end via backprop through the unrolled M-step.

The Twicing Angle: Making Deflation Principled

Classical twicing (Tukey; Milanfar) re-applies the *same* estimator to the residual it just discarded:

$$\hat{\mathbf{x}}_2 = \hat{\mathbf{x}} + D(\mathbf{y} - \hat{\mathbf{x}})$$

Applied to deflation: instead of an arbitrary orthogonalization, generate $\mathbf{x}_t^{(2)}$ by feeding the residual back through the E-step operator itself:

$$r_t = \mathbf{y} - \mathbf{x}_t, \quad \mathbf{x}_t^{(2)} = \text{E-step}(r_t \mid \Theta_t)$$

and recurse ($r_t^{(2)} = \mathbf{y} - \mathbf{x}_t - \mathbf{x}_t^{(2)}$, etc.) for higher-order companions.

Why this is better than ad hoc orthogonalization:

- Not an extra design choice — reuses the existing E-step machinery
- Near-orthogonality to $\mathbf{x}_t$ falls out naturally when the E-step operator behaves like a near-idempotent smoother
- Connects to Lanczos/Krylov-style iteration: repeated operator application to a residual $\Rightarrow$ natural basis for deflation

M-step: Kalofolias-style Log-Barrier

Standard GLASSO-type M-step:

$$\min_{\Theta} \text{tr}(C_t \Theta) - \log \det \Theta + \lambda \|\Theta\|_1$$

$\log \det \Theta$ requires eigendecomposition/inversion — expensive, and ill-suited when C_t is low-rank.

Kalofolias' core idea (decoupled from any Laplacian constraint):

replace the spectral log-det barrier with a cheap, local, per-coordinate barrier. For a *general* (signed, non-Laplacian-constrained) precision matrix, the natural analogue of Kalofolias' degree barrier is the diagonal itself:

$$\min_{\Theta} \text{tr}(C_t \Theta) - \alpha \sum_i \log \Theta_{ii} + \lambda \|\Theta\|_{1, \text{off-diag}}$$

- Retains the inverse-covariance-estimation interpretation — Θ stays a general signed precision matrix, not constrained to be a graph Laplacian
- $\partial/\partial \Theta_{ii} = 1/\Theta_{ii}$ — trivial, cheap to backprop through in an unrolled M-step

Guaranteeing Positive Definiteness Without log det

The diagonal barrier alone does *not* guarantee $\Theta \succ 0$ — it only keeps diagonal entries away from zero.

Fix: pair it with a Gershgorin-style dominance certificate

$$\Theta_{ii} > \sum_{j \neq i} |\Theta_{ij}| \implies \Theta \succ 0$$

(consistent with the Gershgorin-certified ρ^* already used in the CNC sparse graph learning line)

Net effect:

- Fully local/diagonal PD certificate — no global spectral computation
- Barrier's role: *prevent collapse toward the PSD boundary*, not an exact surrogate for $\sum_i \log \lambda_i(\Theta)$
- Cheap enough to sit inside a differentiable, unrolled M-step

Summary

- **E-step diversity:** combine historical iterates (temporal) with twicing-based deflation (spatial, residual-driven) to counter rank-1 C_t
- **Twicing recursion** gives a principled, non-arbitrary recipe for deflated companions $\mathbf{x}_t^{(2)}, \mathbf{x}_t^{(3)}, \dots$
- **M-step:** Kalofolias' log-det $\rightarrow$ log-degree substitution generalizes to a diagonal log-barrier on Θ_{ii} , *without* requiring a Laplacian-constrained Θ
- **PD guarantee:** diagonal barrier + Gershgorin dominance $\Rightarrow$ cheap, differentiable, eigendecomposition-free M-step

Open question: weighting scheme for historical iterates under non-stationary Θ_{t-k} — currently learned via f_ϕ , needs staleness-aware inputs (e.g. iteration index, $\|\Theta_{t-k} - \Theta_t\|$).